

Seismic-AI: Academic Technical Report

Physics-Informed Machine Learning Surrogate Models for Accelerated Multi-Storey Seismic Response Prediction and Design Optimization

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Repository: https://github.com/raghvendra1gdsc-png/seismic_ai • **Verification Suite:** 77 Automated Unit Tests

Abstract: Dynamic time-history analysis (THA) is the gold standard for evaluating structural demands during earthquake excitation. However, its severe computational cost prevents its use in real-time cyber-physical early warning and iterative structural design optimization. This research report presents **Seismic-AI**, an end-to-end computational framework investigating whether physics-informed machine learning surrogates can accurately predict multi-degree-of-freedom (MDOF) multi-storey seismic responses while accelerating simulation time by **over 60,000×**. A from-scratch structural dynamics simulation engine (Rayleigh damping, generalized eigenvalue solver, Newmark- β integration) was constructed and subjected to 24 historical ground-motion records across 40 parametric building configurations. Surrogate models were evaluated across a rigorous **4-Tier Generalization Protocol**, maintaining $R^2 = 0.9510$ under dual-blind testing across unseen earthquakes and unseen buildings, confirmed through closed-loop numerical re-simulation.

1. Theoretical Formulation & Mechanics Engine

Linear and nonlinear MDOF shear-building systems subjected to single-axis horizontal earthquake base motion $a_g(t)$ obey:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{K} \mathbf{u}(t) = -\mathbf{M} \mathbf{r} a_g(t)$$

where $\mathbf{M} = \text{diag}(m_1, \dots, m_N)$, $\mathbf{K}$ is a tridiagonal lateral shear stiffness matrix, and $\mathbf{C} = \alpha \mathbf{M} + \beta \mathbf{K}$ is the classical Rayleigh damping matrix matching target damping ζ at reference modal frequencies. Dynamic responses (interstorey drift ratios $\theta_i(t)$, floor accelerations $\ddot{\mathbf{u}}_i(t)$, and base shear $V_b(t)$) are integrated step-by-step using the Newmark- β Average Acceleration scheme ($\gamma=0.5$, $\beta=0.25$).

2. Physics-Informed Feature Engineering

A parametric suite of 40 multi-storey buildings (3 to 10 storeys, floor masses 60–350 tonnes, stiffnesses 40–400 MN/m) was generated using Latin Hypercube Sampling (LHS) and paired with 24 historical earthquake records (including El Centro 1940, Kobe 1995, Northridge 1994, Chi-Chi 1999). From each simulation, 25 mechanical features were extracted: (1) Modal frequencies & periods (T_1, T_2, T_3); (2) Modal participation & effective mass ($\Gamma_1, M_1^*/M_{\text{tot}}$); (3) Ground-motion Intensity Measures (PGA, PGV, PGD, I_a , D_{5-95}); (4) Spectral response ordinates ($S_a(T_1), S_d(T_1)$); and (5) Dimensionless interaction terms ($S_a(T_1)/\text{PGA}$, Static Drift Proxy).

3. 4-Tier Generalization Protocol & Speedup Results

Protocol Tier	Linear Ridge (R^2)	Random Forest (R^2)	Gradient Boosting (R^2)	PINN Surrogate (R^2)
Tier 1: Random Split	0.9848	0.9340	0.9589	0.9850
Tier 2: Unseen Earthquakes	0.9724	0.8775	0.9463	0.9256
Tier 3: Unseen Buildings	0.9664	0.8865	0.9841	0.9426
Tier 4: Dual-Blind Unseen	0.8748	0.7244	0.9425	0.9510

4. Constrained Structural Design Optimization & Closed-Loop Verification

We formulated lateral stiffness minimization for a 5-storey building under the severe Kobe 1995 record subject to a code drift limit ($\text{PIDR} \leq 1.0\%$):

$$\min \sum k_i \quad \text{s.t.} \quad \text{PIDR}(\mathbf{k}) \leq 0.010, \quad k_1 \geq k_2 \geq \dots \geq k_5$$

Differential Evolution completed 1,050 evaluations in **0.42 s** via the surrogate. Closed-loop numerical re-simulation of the optimal stiffness profile in the high-fidelity Newmark solver yielded an exact true PIDR of **0.702%**, strictly satisfying the 1.0% code requirement with zero AI hallucination.

5. Conclusion & Research Significance

Seismic-AI establishes that physics-informed surrogates achieve $10^3 - 10^5\times$ speedup while maintaining $R^2 > 0.94$ under dual-blind testing. Coupling surrogate exploration with closed-loop numerical verification creates a reliable, mathematically grounded paradigm for computational earthquake engineering.